MAS · 3D Vectors and Systems of Linear Equations — Patterns and Solution Strategies

Based on: marking key + exam reports for 25 past exam questions from 2016–2025. Original questions are in `topics/03_Vectors_3D.pdf`. System-of-equations examples have been checked with sympy. Also references two 2016 sample paper questions (`2016S-A8`, `2016S-A10`; original papers are in `papers/2016Sample_*`). Review status: has undergone one round of independent review and was revised according to feedback (see `Review_log.md`).

Marking-key reminders

- How to write when the three planes do not have a unique solution: the understanding and geometric meaning of "no unique solution" were flagged in the 2018, 2020, 2021, 2022, 2023 and 2024 reports. The 2022 report specifically noted: many people stopped after writing "there are infinitely many solutions" (the 2023 report mentions a similar issue). You must write the solution set—using a parameter, or expressing the remaining variables in terms of one variable (key: "expresses correct relationships between variables"); when the question asks for a vector equation, write $\mathbf{r} = \mathbf{p} + t\mathbf{d}$ (2023F-Q5(c)). You must also state the geometric meaning.
- "No solution" has two geometric cases, which must be distinguished: there are parallel planes; or the three planes intersect pairwise in three parallel lines (like a triangular prism; 2020F-Q4 is exactly this case; the report says this case was never examined in 2016–2019, and some students were completely unprepared).
- Proving two lines do not intersect: you must write the component equations and show there is no solution; "CAS says no solution" does not count as proof (2021 report).
- Vector notation: distinguish position vectors from direction vectors in a plane (2017 report); reports criticised non-standard vector notation (2018, 2022, 2025), and the keys for 2020A-Q20 and 2025A-Q18 have marks tied to correct notation. (Editor's suggestion: add a tilde or bold to vectors, and do not mix up dot product and cross product symbols.)
- Errors in cross product calculations are a common mark loss in CF (2020 report).

Toolkit

- Line: $\mathbf{r} = \mathbf{a} + \lambda\mathbf{d}$; plane: $\mathbf{r}\cdot\mathbf{n} = \mathbf{p}\cdot\mathbf{n}$ ($\mathbf{p}$ is a point on the plane) $\Leftrightarrow ax + by + cz = k$, where $\mathbf{n} = (a, b, c)$; parametric form $\mathbf{r} = \mathbf{a} + \lambda\mathbf{u} + \mu\mathbf{v}$, where $\mathbf{n} = \mathbf{u} \times \mathbf{v}$.
- Sphere: $|\mathbf{r} - \mathbf{c}| = R \Leftrightarrow (x - c_1)^2 + (y - c_2)^2 + (z - c_3)^2 = R^2$ (obtained by completing the square on the general form).
- Angle between a line and a plane: $\cos\varphi = |\mathbf{d}\cdot\mathbf{n}| / (|\mathbf{d}||\mathbf{n}|)$ (taking the absolute value ensures φ is acute), line-plane angle $= 90° - \varphi$; equivalently $\sin\theta = |\mathbf{d}\cdot\mathbf{n}| / (|\mathbf{d}||\mathbf{n}|)$.
- Distance from a point to a plane $= |\mathbf{n}\cdot(\mathbf{P} - \mathbf{A})| / |\mathbf{n}|$, where A is any point on the plane. This formula is not on the formula sheet (2019A-Q19 asks you to prove it); if using it directly in a CA question, it is best to add a one-line derivation, or use the method of "drawing the normal through the point and finding the intersection" (2018A-Q17(c)); volume of a parallelepiped $= |\mathbf{c}\cdot(\mathbf{a} \times \mathbf{b})|$.

Pattern 1: Solving systems of linear equations and discussing the nature of solutions ★★ It has appeared in CF in 7 of the last 10 years (also in CA in 2021)

Standard procedure

1. Elimination: use two pairs of equations to eliminate the same variable (1st mark in the key), solve for the first variable (2nd mark), then substitute back to find the other two (3rd mark).
2. With parameter k: reduce to (coefficient containing k)·z = (constant containing k), e.g. $(k - 2)z = c - 5$ (2025F-Q4):
    - $k - 2 \neq 0 \Rightarrow$ unique solution (c can be any real number) $\Rightarrow$ the three planes intersect at a point;

- $k - 2 = 0$ and $c - 5 \neq 0 \Rightarrow$ no solution;
- $k - 2 = 0$ and $c - 5 = 0 \Rightarrow$ infinitely many solutions $\Rightarrow$ they intersect in a line.

3. When there are infinitely many solutions, write the line (2022F-Q5): let $z = t \Rightarrow y = t - 1$, $x = (11 - 3t)/2 \Rightarrow r = (11/2, -1, 0) + t(-3/2, 1, 1)$.
4. Determining parallelism: normal vectors are scalar multiples of each other $\Leftrightarrow$ planes are parallel. Check the normal vectors first, then decide whether it is "parallel planes causing no solution" or "three parallel lines of intersection (prism)".

Geometric interpretation template

Algebraic result	Geometric meaning
Unique solution	three planes intersect at a single point
Infinitely many solutions (with one parameter)	three planes intersect in a common line (two of the planes may coincide, e.g. 2021A-Q14, the key requires both points to be written)
Infinitely many solutions (with two parameters)	three planes coincide (the solution set is a plane)
No solution, some normal vectors are scalar multiples	at least two planes are parallel (and distinct)
No solution, normal vectors are pairwise non-parallel	planes intersect pairwise in three parallel lines (triangular prism)

Past questions: 2016F-Q6, 2016S-A8, 2018F-Q2, 2020F-Q4, 2021A-Q14, 2022F-Q5, 2023F-Q5, 2024F-Q6, 2025F-Q4.

Pattern 2: Equations of planes (finding normal vectors)

Approach

- Given parametric form: $n = u \times v$ (write each component of the cross product clearly), then substitute a point to find d (2020F-Q2, 2018A-Q17(b)).
- Through three points: cross product of two edge vectors (2022A-Q19(a)), or set $ax + by + cz = d$ and substitute the three points to solve.
- Given $z = 2x + y + 4 \Rightarrow$ rewrite as $2x + y - z = -4 \Rightarrow n = (2, 1, -1)$ (2019A-Q16(a), 2023A-Q14(a)).
- Contains a line and is perpendicular to another plane $\Rightarrow n_2 = d_L \times n_1$ (2019A-Q16(c), 2023A-Q14(d)).
- Contains two intersecting lines $\Rightarrow n = d_1 \times d_2$ (2025A-Q11(b)).

Past questions: 2016F-Q7(b), 2017F-Q7(a)(ii), 2018A-Q17, 2019A-Q16, 2020F-Q2, 2022A-Q19(a), 2023A-Q14, 2025A-Q11.

Pattern 3: Intersections, angles, closest points

Intersection of a line and a plane: substitute $r = a + \lambda d$ into the plane equation $\Rightarrow$ solve for $\lambda \Rightarrow$ find the point (3 marks: substitution, solving λ, writing the point). Line and sphere: after substitution you get a quadratic in λ. Two solutions $\Rightarrow$ two intersections; one repeated root $\Rightarrow$ tangent; no real roots $\Rightarrow$ no intersection (2019A-Q16(d), 2023A-Q9(b)). Alternative method: compare the distance from the centre of the sphere to the line with the radius. Two lines: set the components equal to get three equations; use two of them to solve for λ, μ, then substitute into the third to check. If it is not satisfied $\Rightarrow$ they do not intersect (this is the "proof" required in 2021A-Q16(c)). Closest point on a plane to a point L: construct the line $L + \lambda n$ through L along the normal vector, and find its intersection with the plane (2022A-Q19(b)). Angle between a line and a plane: $\cos \varphi = |d \cdot n| / (|d||n|)$, required angle $= 90° - \varphi$ (2023A-Q14(c), 2024A-Q16(b) angle of descent).

Past questions: 2016A-Q20(a), 2019A-Q16(b)(d), 2021A-Q16(c), 2022A-Q19(b), 2023A-Q9, 2023A-Q14(c), 2025A-Q11(a).

Pattern 4: Spheres

Approach

- From endpoints of a diameter: centre = midpoint (report: many people calculated the position vector incorrectly), radius = |AB|/2 (2016F-Q7(a)).
- General form → standard form: complete the square. Example: $x^2 + y^2 + z^2 - 4x + 2y - 6z + 5 = 0 \Rightarrow (x - 2)^2 + (y + 1)^2 + (z - 3)^2 = 9 \Rightarrow |r - (2, -1, 3)| = 3$ (2023A-Q9(a)).
- Tangent to a plane: radius = distance from the centre of the sphere to the plane; or draw the normal through the centre and find its intersection with the plane (2018A-Q17(c)).
- Passes through all vertices of a rectangular prism: the centre is the midpoint of a space diagonal (2021A-Q16(b)).

Pattern 5: Vector proofs (appears in both CF and CA)

Standard method

1. Use the given basis vectors (a, b, e) to express the required vectors: PQ = OQ − OP = (a + kb) − ka.
2. Calculate the dot product and expand (the key gives one mark for "expanding to get 4 terms").
3. Use the given conditions: square ⇒ a·b = 0 and |a| = |b|; midpoint ⇒ ½(a + b).
4. Draw the conclusion: PQ·QR = 0 ⇒ ∠PQR = 90°.

Past questions: 2016S-A10 (proof of the point-to-line distance formula), 2019A-Q19 (distance between parallel planes), 2020A-Q20 (right angle in a square), 2018F-Q8(b) (why c·(a × b) is the volume: |a × b| is the base area, and the projection of c onto the normal is the height), 2025A-Q18 (pyramid).

Pattern 6: Contextual applications (laser reflection, steepest descent on a ramp, drone approaching a tower)

- Reflection (2016A-Q20(b)): from B to S unit vector + from B to R unit vector = k·n (diagonal of a rhombus). Equivalent form: đ_out − đ_in ∥ n. Note that the key text says "direction of SB", but what is actually substituted is the B→S direction; adding using the S→B direction gives a vector parallel to the mirror plane, not parallel to n.
- Steepest descent direction on a ramp: lies in the ramp plane and is perpendicular to the horizontal line AB (2022A-Q19(c)).
- Motion and distance fall into two categories (see the "Vector Calculus" notes for details):
- (i) Same time parameter t: for the closest distance use r(t)·(relative velocity) = 0; for how long |r(t)| < R lasts, solve an inequality (2017A-Q14(c)).
- (ii) Shortest distance between two lines in space: use two independent parameters λ, μ and set $PQ \cdot d_1 = PQ \cdot d_2 = 0$ (2024A-Q16(c): Ben can adjust speed and departure time; another solution method for 2021A-Q16(c)).